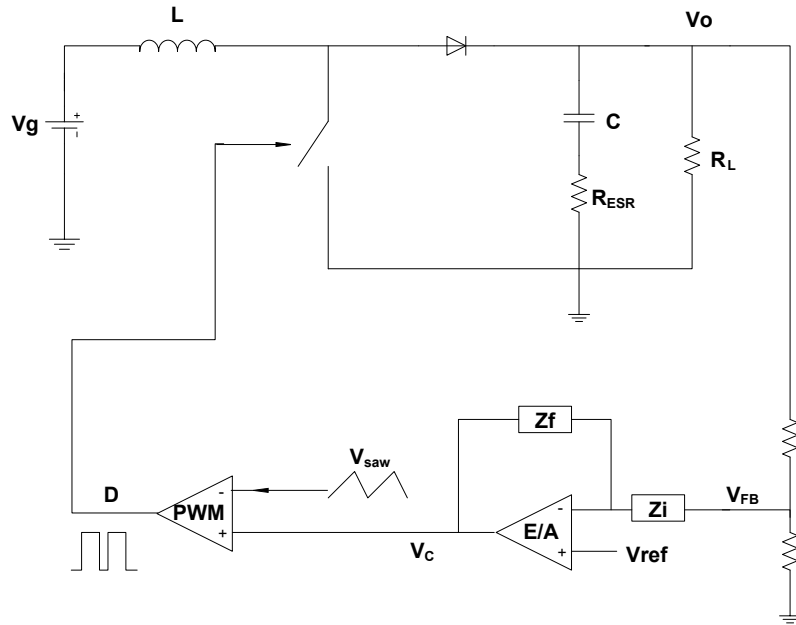


Homework 3

- 1. (10%) Group Discussion: PE-4-P. 13
- 2. (10%) Group Discussion: PE-4-P. 16
- 3. (10%) Group Discussion: PE-5-P. 37



P.13 Which PWM node shall V_c connect to? How to know V_c value at steady-state?



Boost converters: $\frac{V_o}{V_g} = \frac{1}{1-D}$

$D \uparrow$ then V_o goes $\uparrow$

Check: When V_o goes high

$\downarrow$
 V_c goes low

$\downarrow$
 D goes low

$\downarrow$
 V_o goes low

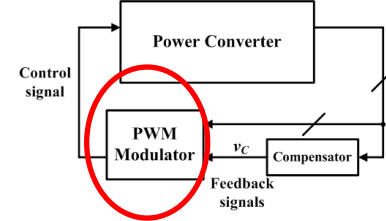
$\downarrow$
Stable

E/A 's output should be connected to Comparator's () node

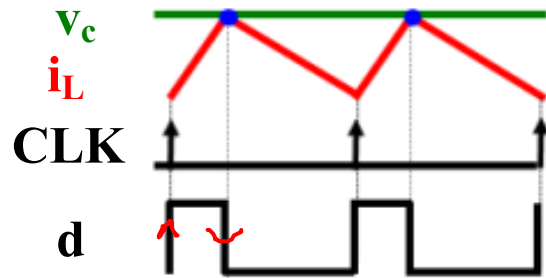
Group Q: Which PWM node shall V_c connect to?

How to know V_c value at steady-state?

PE-4- P. 16: Explain the control laws and why modulation waveform look like this

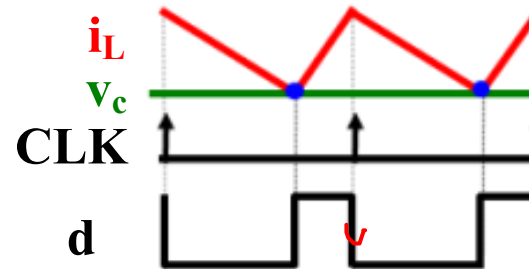


Peak control



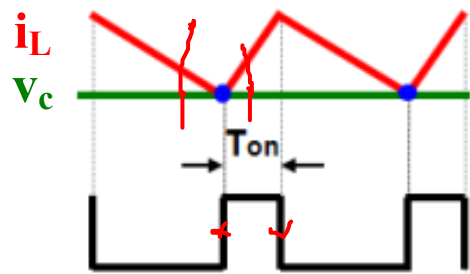
(a)

Valley control

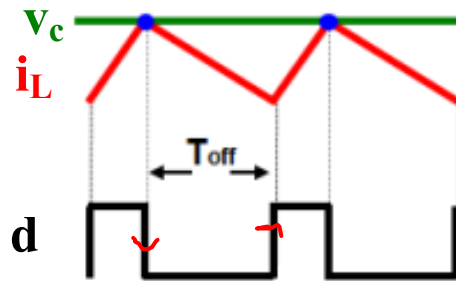


(b)

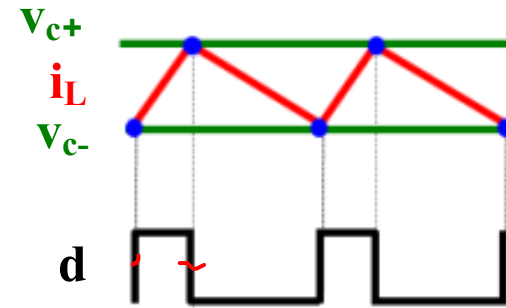
Constant f_{sw}



Constant on-time control



Constant off-time control



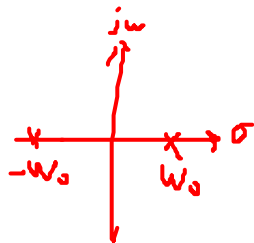
Hysteretic control

Group Q:
Modulation
waveform
Control law



PE-5- P. 37: Problem set

1. Draw the Bode plot (gain and phase) of a transfer function with (1) single LHP pole, (2) LHP zero, (3) RHP zero, (4) a complex pole
2. Draw the Bode plot of transfer function with a pole at origin, a RHP zero at 1 KHz, a RHP pole at 10 KHz

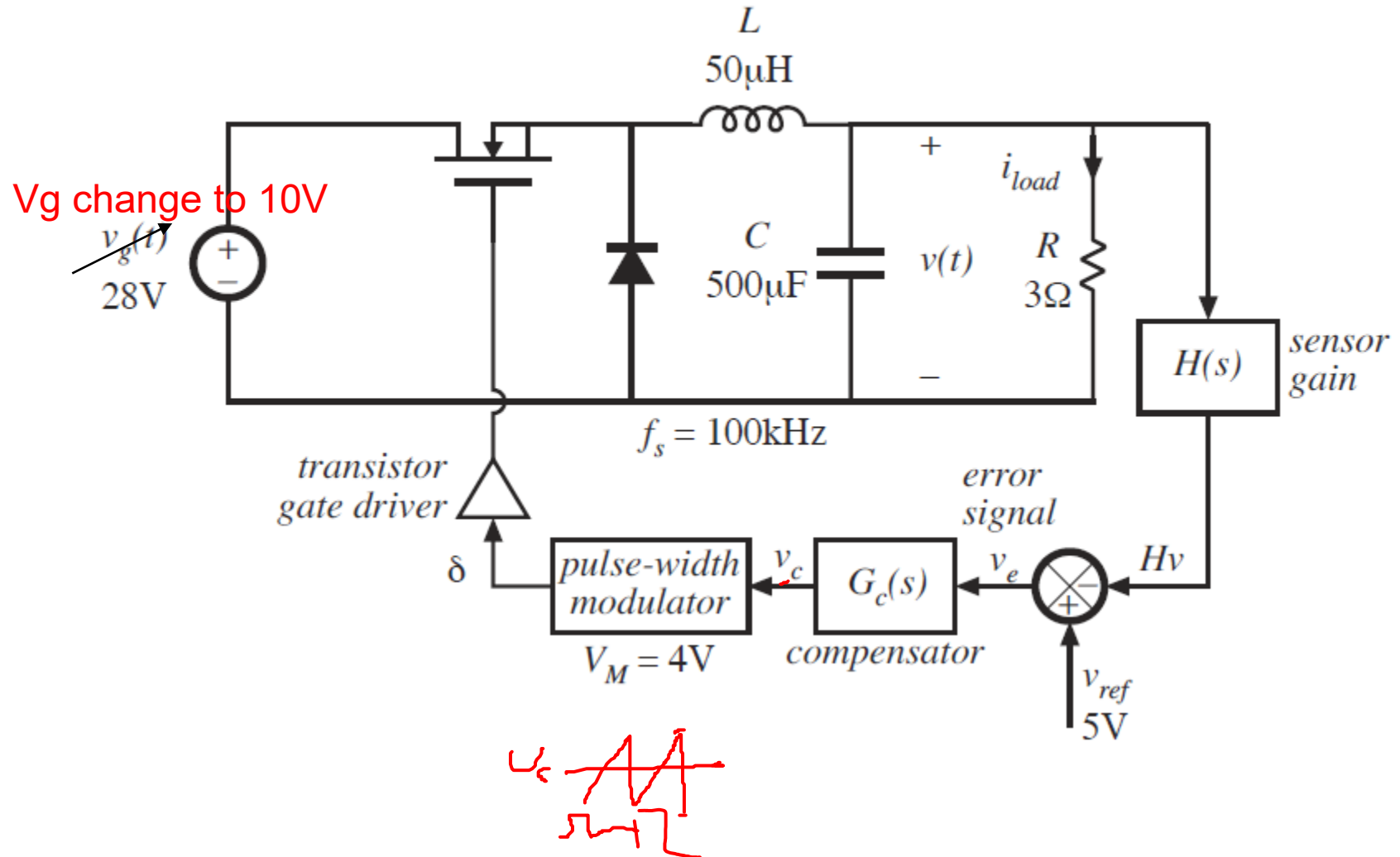


- 4. PEIC 6-p.16: Design a Voltage-Mode Controlled Buck Converter
- Please design and implement a trailing-edge constant-frequency voltage-mode controlled buck converter.
- The spec. follows PEIC 6-p.16, 17 while V_g change to 10V, V_{out} change to 5V, V_{ref} change to 1V, R change to 1 ohm.
The voltage of external ramp is between 1V and 5V.
- Design the compensator according to the uncompensated Loop Gain to achieve high loop gain bandwidth

- (1) Show the implemented close-loop behavior circuit in simulation software. SIMPLIS is recommended. Show the simulated steady-state waveforms of V_{out} and PWM comparator inputs/output signals. (10%)
- (2) Show how you design the compensator transfer function, compensation network values, and loop gain. Use the simulations of loop gain and compensator gain to verify your design. (50%)
- (3) If V_g has 100 mV sin. perturbation at 1 kHz, what is V_o perturbation magnitude before and after loop close? Use simulation to verify your calculation (10%)

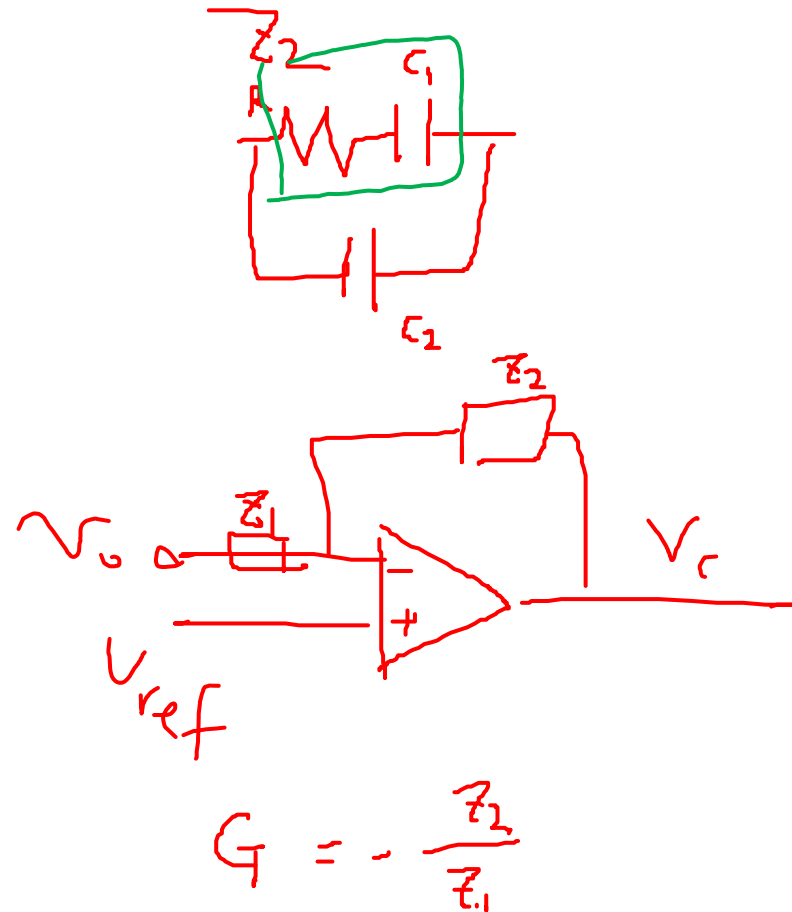
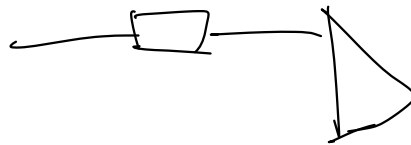
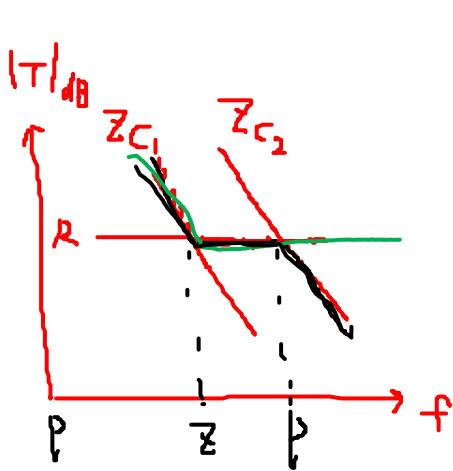


PEIC 6-p.16: Design Example of a Voltage-Mode Controlled Buck Converter



PE-5- P. 39: compensation network design: Derive pole/zero equations of a compensation network

$$Z_{c1}(f=f_z) = R$$



$$-20 \frac{dB}{dec}$$

How to estimate compensation pole/zero freq.?

Fast Analysis of Compensator Pole/Zero Locations

- Using impedance asymptotes in dB axis

- In dB axis: If $Z_1 \gg Z_2$,

$$Z_1 + Z_2 \sim Z_1$$

$$Z_1 // Z_2 \sim Z_2$$

Note:

More accurate when impedances are differ much

- Example:

$$Z_1 = Z_2 \rightarrow \frac{1}{sC_1} = R_2$$

